Priyanjali Patel

Data Scientist | GenAI, LLM & RAG

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience building GenAI applications — Retrieval-Augmented Generation (RAG) pipelines, multi-agent LLM workflows, and LLM-powered multimodal/clinical NLP systems — alongside deep learning work in computer vision (CNN/U-Net segmentation) and sequence modelling (DNN/LSTM), using OpenAI GPT, Gemini, and GROQ-hosted LLMs with LangChain and LangGraph. PhD-trained with 5+ years in data science and 1 year turning multimodal data (text, tabular, time-series, image) into production-ready analytical and AI systems. Experienced integrating LLMs with REST APIs, SQL databases, and FastAPI/Streamlit applications; skilled in prompt engineering, vector search, and multi-agent orchestration, translating complex model outputs into business-ready outcomes for cross-functional, multicultural stakeholders.

WORK EXPERIENCE

MEDITECH INNOVATIONS 08/2025 – Present

Data Science Intern · Remote, Noida West, India

[Data: Multimodal (medical images: MRI, CT, X-ray, ECG, ultrasound, histopathology) | Text (clinical NLP) | Tabular (health records) | Time-series (clinical sequences)]

- Built a Medical Visual Question Answering (VQA) system using the Gemini multimodal model (3.6 Flash) for AI-assisted interpretation across seven clinical imaging modalities; designed prompt-engineered extraction templates that pull diagnoses, symptoms, and treatment plans as structured tables from unstructured clinical notes.
- Built ChatRIA, a Retrieval-Augmented Generation (RAG) assistant for pharmaceutical regulatory Q&A: parsed and chunked FDA drug-label PDFs (LangChain), embedded content with OpenAI embeddings into a FAISS store, and used MMR retrieval with GPT-4o for structured, source-cited answers; deployed via Streamlit.
- Applied ML/DL models (Logistic Regression, DNN, LSTM on sequential clinical time-series) for predictive insights; exposed LLM and analytical outputs via Streamlit dashboards integrated with REST APIs for real-time clinical decision support.

Collaborative Project – ML-based Fraud Detection 11/2025 – 01/2026

[Data: Structured tabular (imbalanced financial transaction records)]

- Built a fraud detection classifier (XGBoost) on imbalanced tabular transaction data; evaluated via ROC-AUC and precision-recall AUC, translating threshold trade-offs into business-aligned risk recommendations.

PERSONAL PROJECTS

Wash-Trading Risk Scoring **Web3 / Crypto Analytics · Uncertainty Quantification**

[Data: On-chain NFT trade history (Dune Analytics, public)]

- Built an NFT wash-trading risk-scoring model on public on-chain data; engineered features from reciprocal trades, tight timing, and counterparty concentration to flag suspicious wallets without ground-truth labels.
- Quantified per-wallet uncertainty via bootstrap-resampled XGBoost (200 resamples, 90% intervals), cross-checked against a label-free Isolation Forest; recovered 83.3% of injected wash-trading rings at high confidence.

github.com/priyanjalipatel/wash-trading-risk-scoring

Glacier Segmentation with U-Net **Deep Learning · Computer Vision · Remote Sensing**

[Data: Synthetic multi-band satellite imagery]

- Extended the U-Net segmentation work from Omdena into a standalone, open-sourced PyTorch pipeline — patch-based dataset generation, multi-band preprocessing, and Dice-loss training — to practise the full deep learning workflow independently, end to end.

github.com/priyanjalipatel/glacier-segmentation-unet

TECHNICAL SKILLS

GENAI, LLM & AGENTIC AI

- LLM APIs: OpenAI GPT, Gemini Pro Vision, GROQ
- LangChain, LangGraph, CrewAI
- Retrieval-Augmented Generation (RAG)
- Vector Search & Embeddings (FAISS)
- Prompt Engineering, Multi-Agent Workflows
- Tool Calling, LLM-based NLP Pipelines
- Hugging Face, Transformer Architectures (built GPT from scratch)

MACHINE LEARNING & DEEP LEARNING

- CNNs, U-Net (Computer Vision / Semantic Segmentation)
- DNN, LSTM (Sequence Modelling)
- XGBoost, Random Forest, Gradient Boosting
- Isolation Forest, Anomaly Detection
- Bootstrap Resampling & Uncertainty Quantification
- PyMC (Bayesian Modelling)

LANGUAGES & BACKEND

- Python (Pandas, NumPy, scikit-learn, PyTorch)
- SQL (Intermediate), R (basic)
- FastAPI; REST API Integration
- PySpark, Databricks

WORK EXPERIENCE (CONT'D)

OMDENA 09/2023 – 08/2025

Machine Learning Engineer (Volunteer) · Remote, Palo Alto, USA

[Data: Image (multi-band Landsat satellite imagery) | Text (financial documents, PDFs, queries) | Structured relational (SQL)]

- **Designed, trained, and evaluated a U-Net convolutional neural network (PyTorch) for pixel-wise glacier segmentation on multi-band Landsat satellite imagery** — building the deep learning pipeline end-to-end (patch-based dataset generation, multi-band preprocessing, Dice loss to handle severe foreground/background class imbalance) and reaching a validation Dice coefficient of ~0.82.
- Designed and built a multi-agent Retrieval-Augmented Generation (RAG) + SQL chatbot using GROQ-hosted LLMs for an AI banking platform, orchestrating retrieval over unstructured financial text with tool-calling into structured transactional data; led the chatbot team's agent-workflow design, contributed to Fraud Model integration, and delivered data-driven insights through automated analytics workflows.

DATA GLACIER 04/2025 – 05/2025

Data Scientist Intern · Remote, Prayagraj, India

[Data: Structured tabular (customer banking records — demographic & behavioural features)]

- Built classification models (Logistic Regression, Random Forest, Gradient Boosting) predicting term deposit subscriptions for a BFSI marketing campaign; achieved ROC-AUC of 0.92 via SMOTE class balancing and validated outputs through confusion-matrix / classification-report analysis.

VIRIDIEN (formerly CGG) + S2DS Fellowship 02/2024 – 03/2024

Data Engineer · Remote, London, UK

[Data: Text (unstructured source code & technical documentation for LLM-based refactoring)]

- Selected for the competitive S2DS Fellowship; built scalable ETL pipelines to extract and transform unstructured code and technical documentation using LangChain and OpenAI GPT APIs for LLM-based automated code refactoring in an Agile team for a global HPC leader.

UNIVERSIDAD DE CHILE 03/2019 – 12/2024

Doctoral Researcher · Santiago, Chile

[Data: Time-series (large-scale multi-wavelength photometric survey, ~7,000 astronomical objects)]

- Calculated quasar optical variability using C code (Mexican Hat code), identifying and fixing a critical error to enhance short-timescale variability detection.
- Conducted analysis of quasar optical variability using Python and advanced statistical techniques: retrieved and processed large-scale time-series data (light curves) for ~7,000 quasars via APIs from the Zwicky Transient Facility (ZTF) survey; applied statistical modelling (**bootstrapping with 1,000 samples**, **regression analysis**) and Bayesian inference / Markov Chain Monte Carlo (MCMC) techniques for parameter estimation, uncertainty quantification, and extracting insights from complex, noisy time-series datasets.
- Developed Python-based data analysis pipelines for the ALeRCE ML collaboration, improving processing efficiency by 15%; performed quality filtering of time-series data for ML-ready astronomical datasets.
- Contributed as a co-author to two scientific papers, involving text review and preliminary investigations of light curves.
- Delivered presentations on research findings at scientific conferences, both locally and internationally.

UNIVERSIDAD DE CHILE 03/2020 – 01/2023

Teaching Assistant · Santiago, Chile

- Graded astronomy exams for five semesters and developed well-structured exams for one semester.

PUBLICATIONS

P. Patel et al., Probing the rest-frame wavelength dependence of quasar variability, A&A, 695, A162 (2025).

P. Arévalo et al., Universal power spectrum of quasars in optical wavelengths, A&A, 684, A133 (2024).

P. Arévalo et al., Optical variability in quasars: scalings with BH mass and Eddington ratio, MNRAS, 526, 6078 (2023).

CLOUD & MLOPS

- AWS (S3, EC2, SageMaker)
- Docker
- Git, Agile Methodology

DATA ANALYSIS & STATISTICS

- EDA, Data Cleaning & Wrangling
- Statistical Modelling
- Hypothesis Testing, A/B Testing
- Regression & Time-Series Analysis
- Bayesian Modelling, MCMC
- On-Chain / Web3 Data (Dune Analytics)

VISUALIZATION & REPORTING

- matplotlib, seaborn, Plotly
- Streamlit (interactive AI apps)
- Business Dashboard & KPI Reporting

SOFT SKILLS

- Stakeholder Communication
- Cross-functional Collaboration
- Data-Driven Decision Making

EDUCATION

Ph.D. in Astrophysics
Universidad de Chile
Santiago, Chile | 2019–2024
ANID National Fellowship Recipient

M.Sc. in Physics
Pt. Ravishankar Shukla Univ.
Raipur, India | 2015–2017

B.Sc. Physics, CS & Math
St. Thomas College
Bhilai, India | 2012–2015

AWARDS

ANID PhD Fellowship
National Research & Development Agency, Chile | 2019–2024

LANGUAGES

English (Fluent)
Hindi (Native)
Spanish (Intermediate)